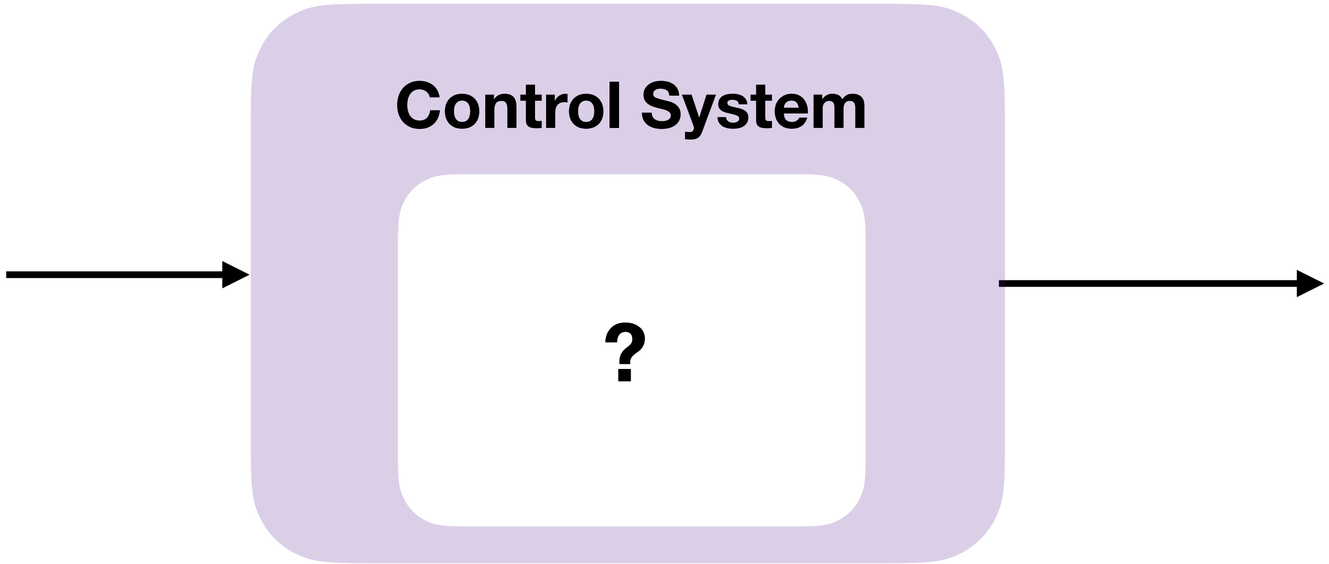
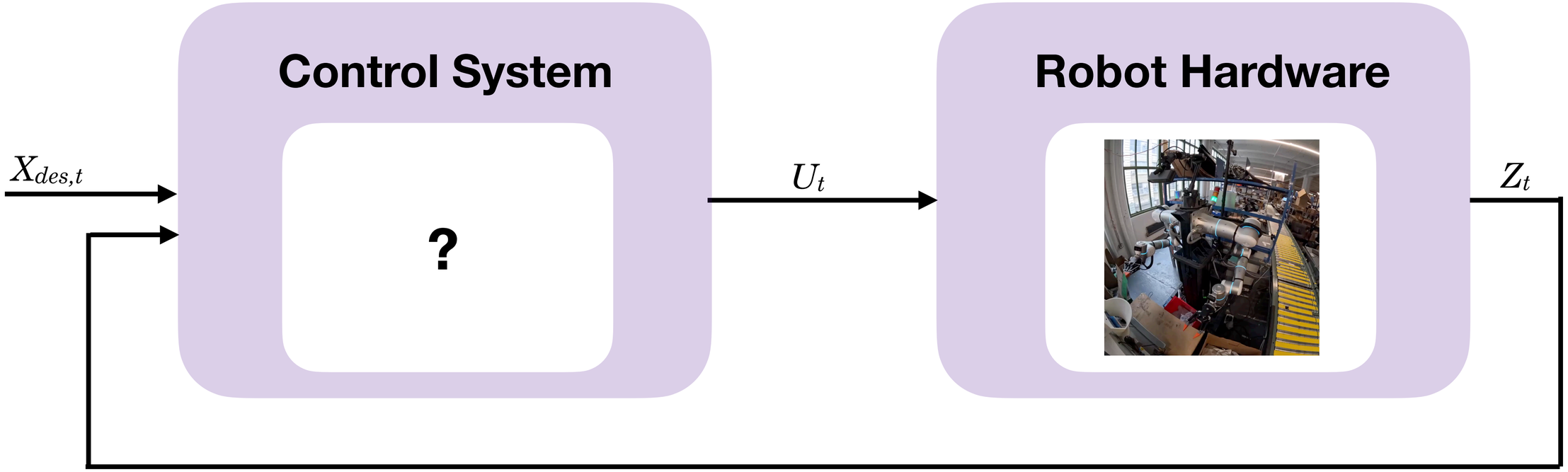
ROB-UY 3303

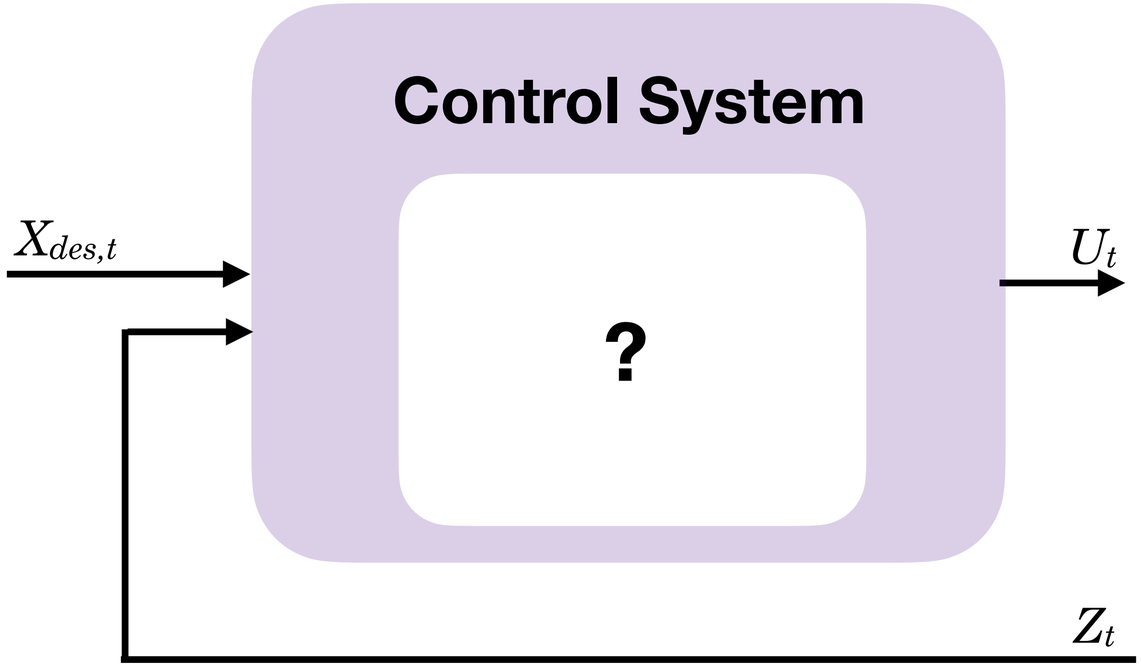
Robot Motion & Planning

Lecture 02A - Robot Control Architectures



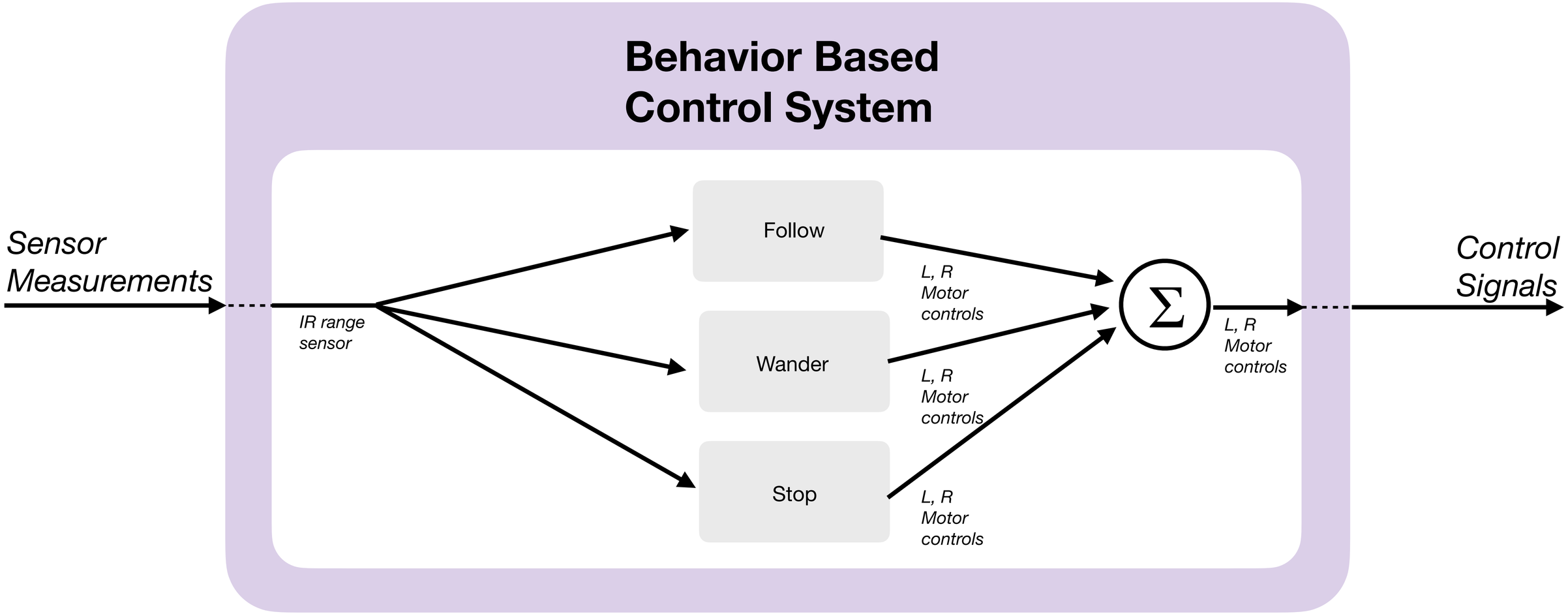
CLOSED LOOP CONTROL

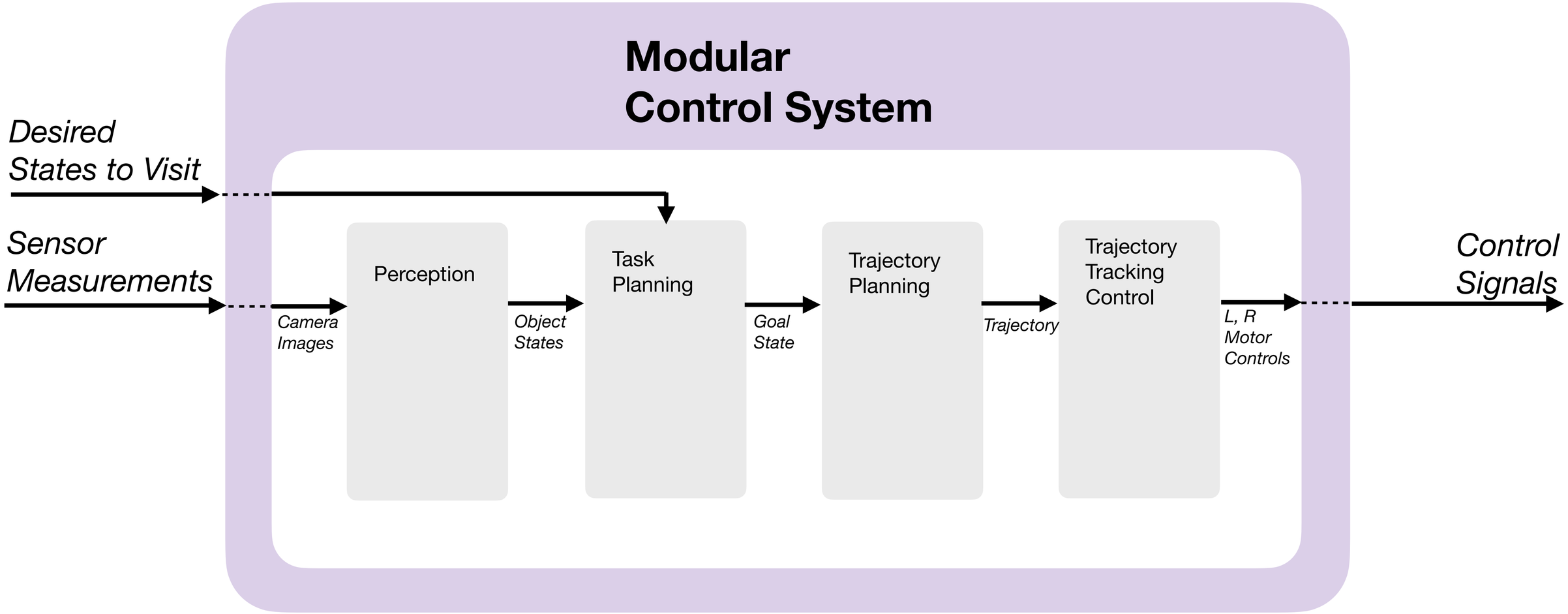




Control System Architectures

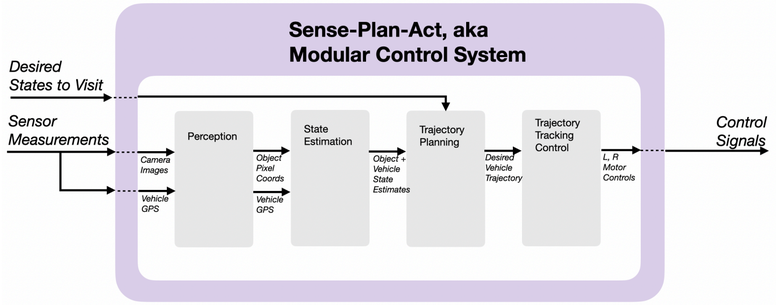
1. Behavior Based
2. Sense-Plan-Act (aka Modular)
3. Hybrid Architectures
4. End-to-End (ML)


Behavior Based
Control System
Sensor
Measurements
IR range
sensor
Follow
Wander
Stop
L, R
Motor
controls
L, R
Motor
controls
L, R
Motor
controls
Σ
L, R
Motor
controls
Control
Signals


Modular
Control System
Desired
States to Visit
Sensor
Measurements
Perception
Task
Planning
Trajectory
Planning
Trajectory
Tracking
Control
Control
Signals
Camera
Images
Object
States
Goal
State
Trajectory
L, R
Motor
Controls

Modular Control Systems

- Are characterized by their “division of labor”

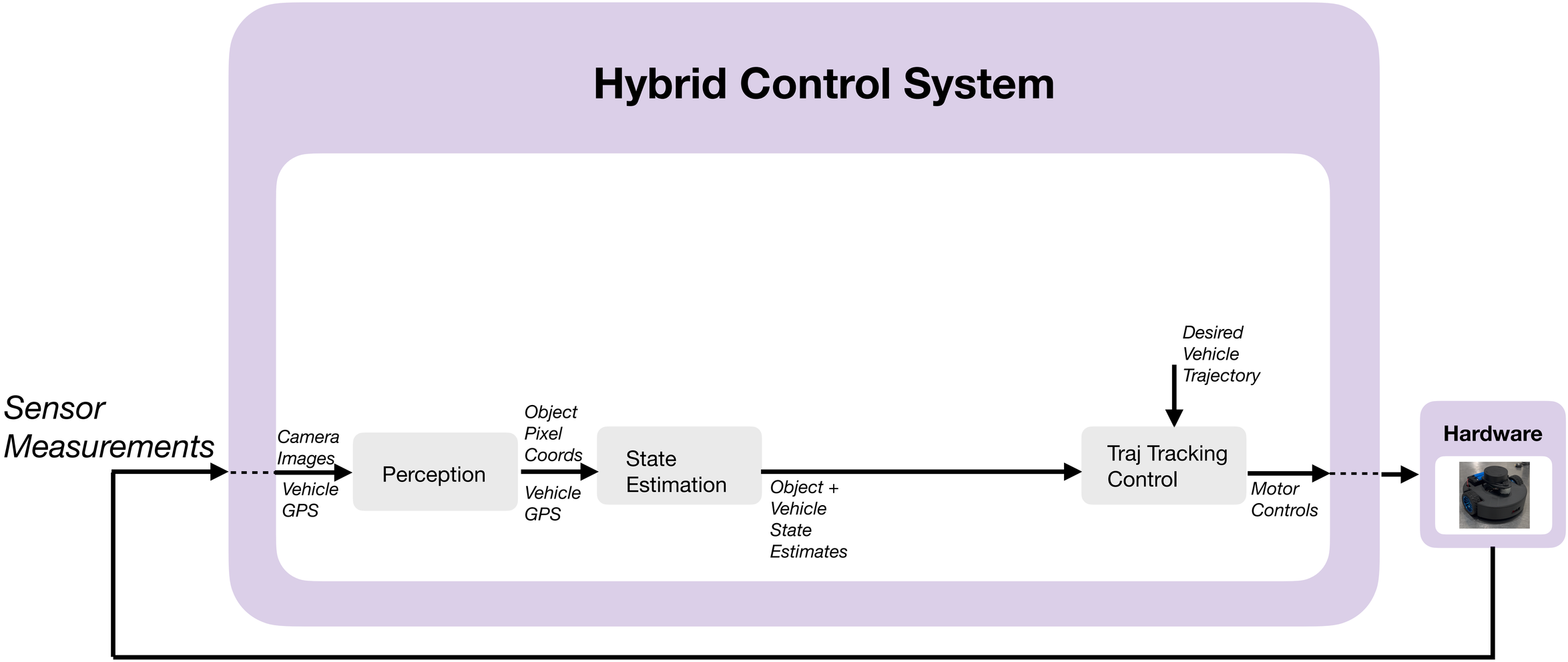


Hybrid Control Systems

- Also characterized by their “division of labor”
- But, have many control loops that run:
    - Asynchronously
    - At different rates

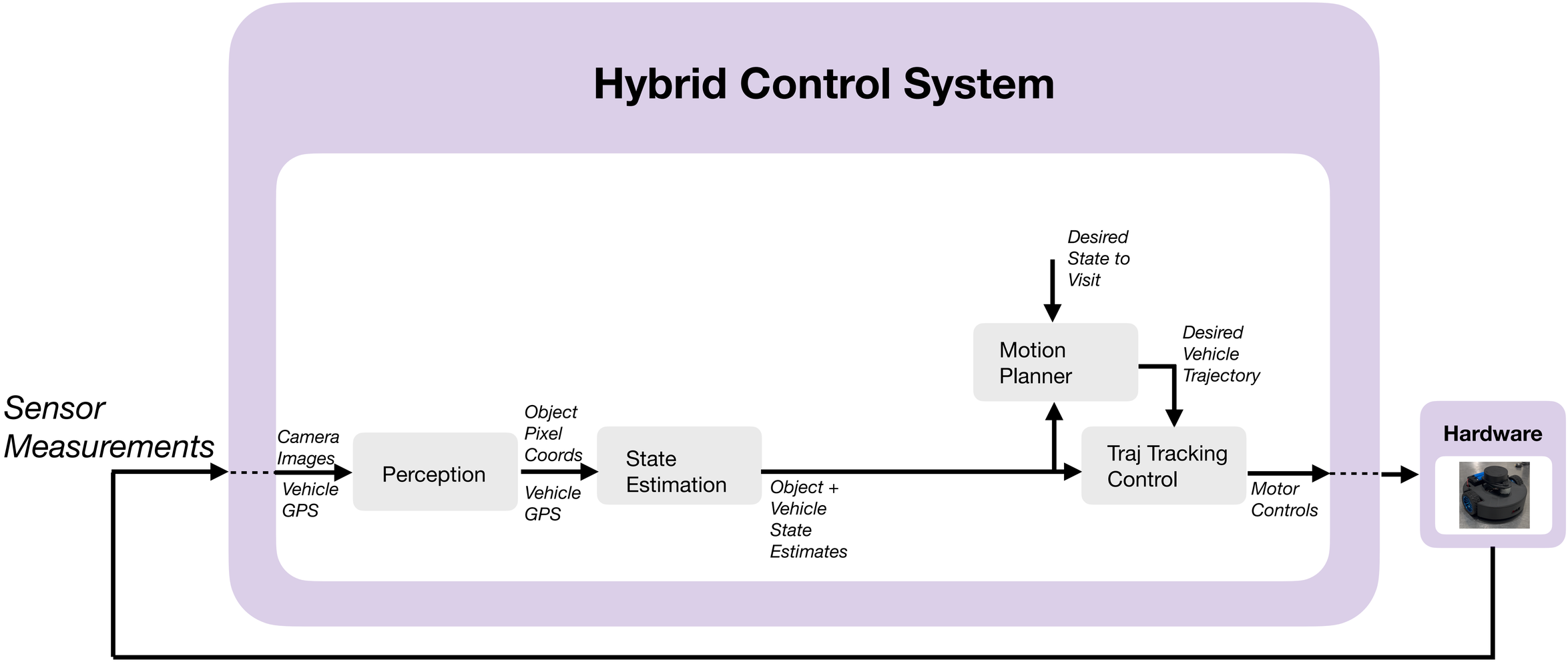
ROB-UY 3303

Robot Motion & Planning



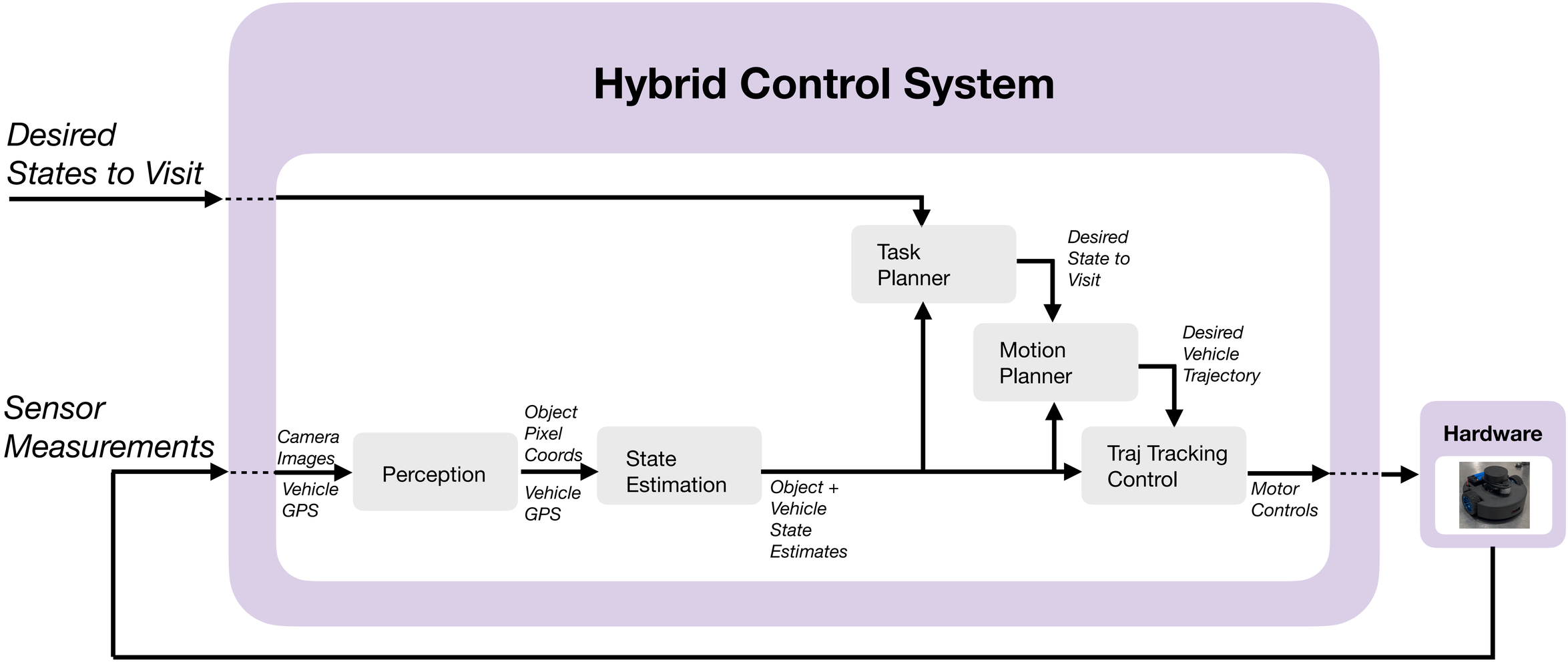
ROB-UY 3303

Robot Motion & Planning



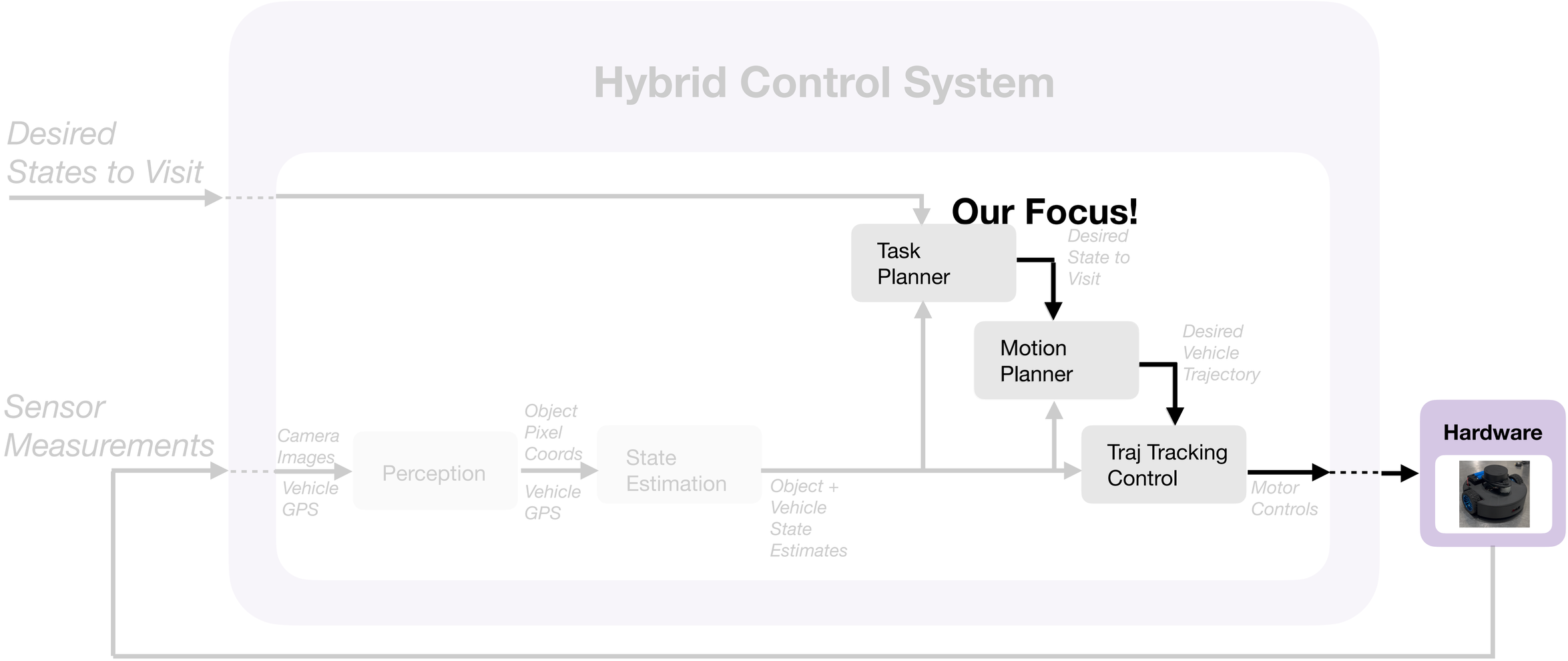
ROB-UY 3303

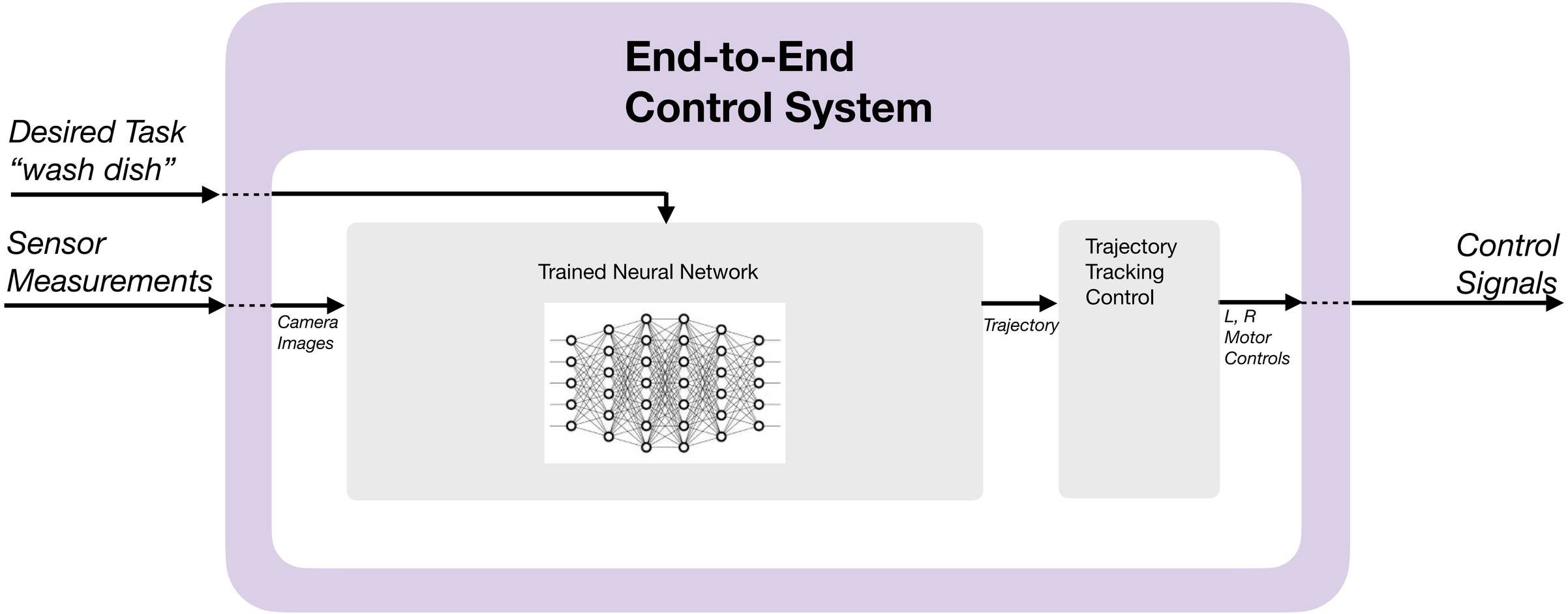
Robot Motion & Planning



ROB-UY 3303

Robot Motion & Planning




End-to-End
Control System
Desired Task
“wash dish”
Sensor
Measurements
Camera
Images
Trained Neural Network
Trajectory
Trajectory
Tracking
Control
L, R
Motor
Controls
Control
Signals